Robotics And Autonomous Systems

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Module Code: CS3S662

Panopto Link:

https://southwales.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=ccb10823-203d-4e76-b094-b437010de220

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Introduction

Early research in robotics showed that meaningful responses to environmental stimuli can emerge from simple connections between sensors and motors. Braitenberg demonstrated that even basic sensor driven vehicles can follow physical traces and respond to encoded information in their surroundings (Smart, 2020). This idea provides the conceptual basis for the present project.

In this project, a Lego EV3 robot was programmed to read a message encoded in Morse code along a printed paper track. A colour sensor detects transitions between red and white segments, while a timer measures the duration of each mark and gap to distinguish between dots, dashes, and character boundaries. The decoded message is displayed progressively on the EV3 screen during movement.

This report presents the system design, implementation process, and testing performed on the physical robot. It also discusses practical limitations observed during operation and briefly considers the role of simulation in supporting development.

System Overview

The system consists of a Lego EV3 robot programmed to decode a Morse code message encoded as coloured bars on a printed paper track.

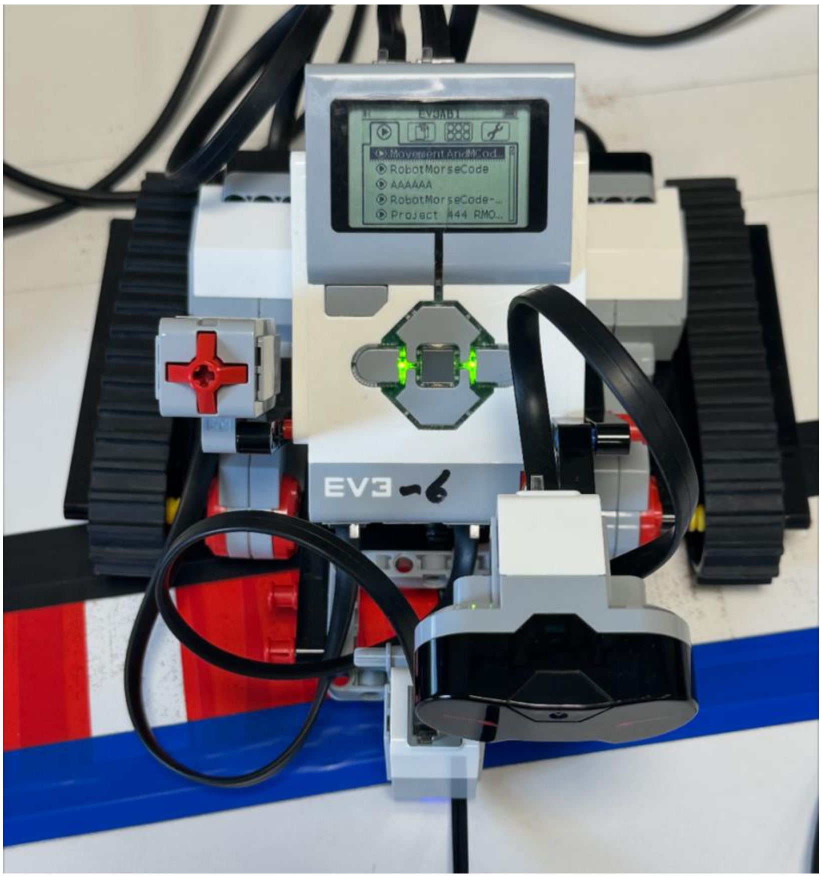

Figure 1: EV3 robot configuration used for Morse code decoding along the printed track

The robot moves forward at a constant speed while a colour sensor mounted beneath the chassis monitors transitions between red and white segments. Red segments represent Morse marks, while white segments represent the gaps between them.

Operation begins when the centre button on the EV3 brick is pressed. Once activated, the robot measures the duration of each red segment to determine whether it represents a dot or a dash. White gaps are also timed to identify boundaries between symbols and characters. Detected symbols are accumulated into a sequence and matched against a predefined Morse alphabet to produce the corresponding character. Each decoded character is displayed on the EV3 screen as the robot progresses along the track.

An ultrasonic sensor mounted at the front of the robot monitors the area ahead during movement. If an obstacle is detected within a defined distance, the robot stops immediately to prevent unintended movement beyond the reading area.

Hardware Configuration

The robot used in this project followed the standard Lego EV3 laboratory configuration provided for the module. No physical modifications were required, as the platform was already suitable for colour sensing and track based navigation.
Two large motors connected to ports B and C drove the robot forward along the printed track. A consistent movement speed was maintained because the decoding process depended on timing the duration of coloured segments to distinguish between dots and dashes.

A colour sensor connected to port 3 was mounted facing the track surface and detected transitions between red and white segments representing Morse marks and gaps.

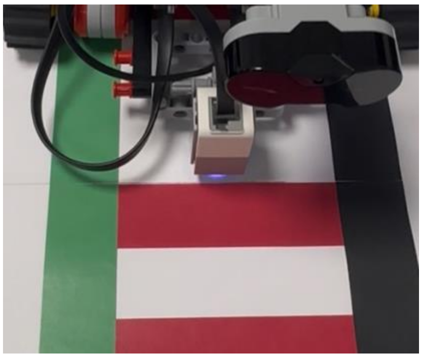

Figure 2: Placement of the colour sensor above the printed Morse code track

An ultrasonic sensor connected to port 4 monitored the area in front of the robot. If an obstacle was detected within a defined distance, the robot stopped to prevent unintended movement beyond the reading area.

Program Design

The program was structured around four main stages: sensing, timing, classification, and translation. This sequence allowed the robot to process coloured segments continuously while moving along the printed track and convert them into readable characters in real time.

Execution begins when the centre button on the EV3 brick is pressed. The robot then moves forward at a constant speed while the colour sensor monitors transitions between red and white segments.
When a red segment is detected, a timer measures its duration and compares it with a predefined threshold to classify the signal as either a dot or a dash.

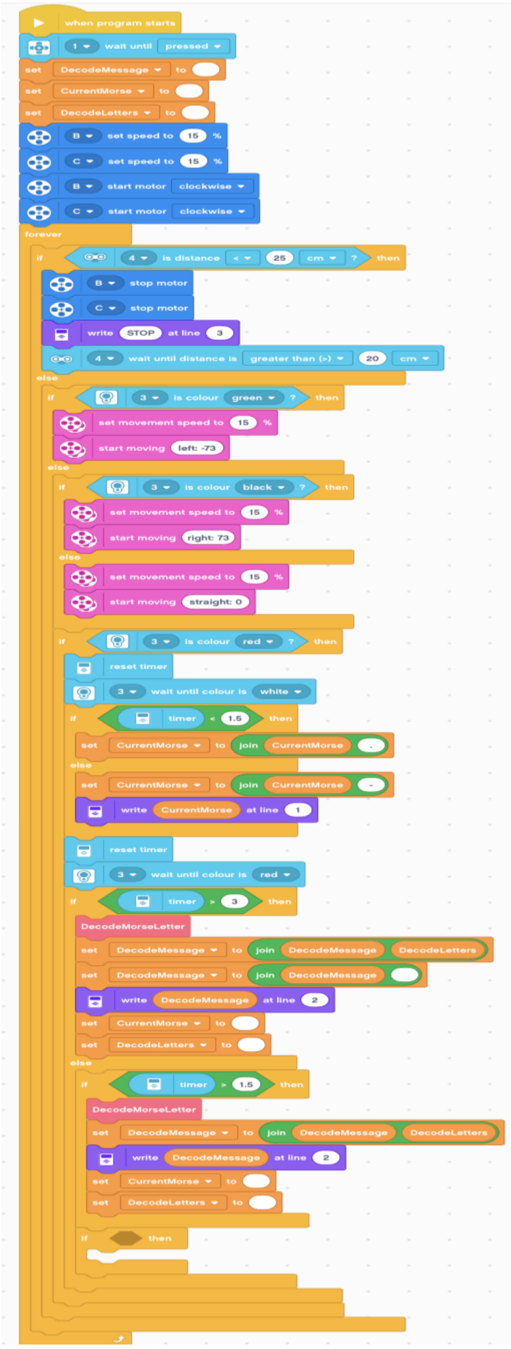

Figure 3: Program structure for timing based Morse code decoding

White gaps between segments are also timed to identify boundaries between symbols and characters. Longer gaps are interpreted as word separation and result in a space being added between decoded words.

Detected symbols are stored temporarily and matched against a predefined Morse alphabet. Once a match is found, the corresponding character is displayed on the EV3 screen as decoding continues along the track.

During operation, the ultrasonic sensor monitors the area ahead of the robot and stops movement if an obstacle is detected within a defined distance.

Testing Strategy

Testing was carried out incrementally, beginning with individual component checks and progressing towards full message decoding on the physical track.

The first stage confirmed that the colour sensor could reliably detect transitions between red and white segments under laboratory lighting conditions. Threshold values were adjusted where necessary to improve detection stability.

The second stage focused on the timing mechanism used to classify dots and dashes. Segment durations were monitored during movement to confirm that short and long marks could be separated correctly using the defined threshold values.

Spacing detection was then tested by observing white gaps of different lengths. This confirmed that the program correctly identified boundaries between symbols and characters and triggered translation at the appropriate time.

Finally, full track tests were performed using complete Morse encoded patterns. The decoded output displayed on the EV3 screen was compared with the expected message to confirm overall decoding accuracy.

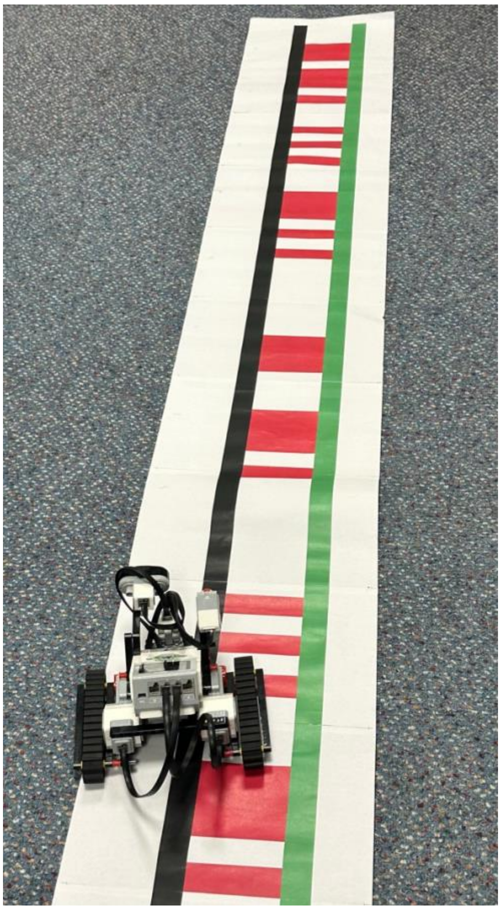

Figure 4: Printed Morse code track used during testing

Physical Constraints and Implementation Challenges

Several physical limitations of the EV3 platform affected decoding reliability while the robot operated along the printed track.

Maintaining a constant movement speed was one of the main challenges. Because symbol classification depended on measuring the duration of red segments, variations in motor speed could influence the distinction between dots and dashes. This effect was reduced by selecting a stable speed during operation.

Colour sensor sensitivity also affected performance. Changes in laboratory lighting conditions sometimes produced inconsistent readings when detecting transitions between red and white segments. Adjustments to sensor position and detection thresholds improved reliability during testing.

Robot alignment relative to the track influenced measurement accuracy as well. If the sensor shifted away from the centre of the coloured pattern, readings became less stable. Careful positioning at the start of each run helped reduce misclassification.

The ultrasonic sensor required threshold calibration to prevent unnecessary stopping during normal movement. Although it provided an important safety function, its detection distance needed adjustment to avoid false triggers.

Simulation vs Physical Robot Behaviour

Simulation environments provide a controlled setting for developing and testing robotic behaviour before applying it to physical hardware. In this project, simulation supported early testing of movement logic, colour detection, and timing based decoding without environmental variation.

A key difference between simulation and real operation was sensor consistency. In simulation, colour readings remain stable, while on the physical robot they were affected by lighting conditions and small positioning changes along the track. This influenced the reliability of detecting transitions between red and white segments.

Movement behaviour also differed between simulated and physical operation. Simulated motion is typically precise, whereas the real robot showed small variations caused by motor response and surface conditions. Because Morse decoding depended on timing measurements, these variations sometimes affected symbol classification.

Simulation models also cannot fully represent real physical interaction, meaning behaviours developed in simulation may require adjustment when transferred to hardware (Chukwurah, Adebayo and Ajayi, 2024).

Despite these limitations, simulation remained useful for verifying program logic before physical testing and helped reduce implementation errors during development.

Simulation Usefulness

Simulation tools supported development by providing a controlled environment in which movement logic and decoding behaviour could be tested before running the program on physical hardware. This allowed adjustments to be made quickly without repeated setup on the real robot.

One practical advantage of simulation was the reduction in physical testing time. Different movement strategies and detection conditions could be evaluated in the virtual environment, helping identify logical errors at an early stage and reducing issues during hardware testing.

Simulation also provided a safer and more efficient development approach compared with repeated physical trials. It allowed behaviours to be explored without risk to hardware and without dependence on equipment availability, which is why simulation is widely used in robotics development workflows (Afzal et al., 2020).

Verification

Verification was carried out to confirm that the timing based decoding logic produced consistent and accurate outputs while the robot moved along the printed track.

Initial checks confirmed that the colour sensor reliably detected transitions between red and white segments, allowing the robot to identify the start and end of Morse marks during movement.

The timing mechanism used to distinguish between dots and dashes was verified through repeated test runs across segments of different lengths. Results showed that the selected threshold provided stable symbol classification.

Character boundary detection was also confirmed by testing white gaps of different durations. Longer gaps correctly triggered translation of accumulated symbols into letters.

Finally, complete Morse encoded sequences were tested and the decoded characters displayed on the EV3 screen matched the expected message, confirming correct operation of the decoding process under controlled conditions.

Result

The implemented system successfully decoded a Morse code message from the printed track during physical testing. During laboratory testing, the robot successfully decoded the message RAS AT USW, which was displayed on the EV3 screen during operation.

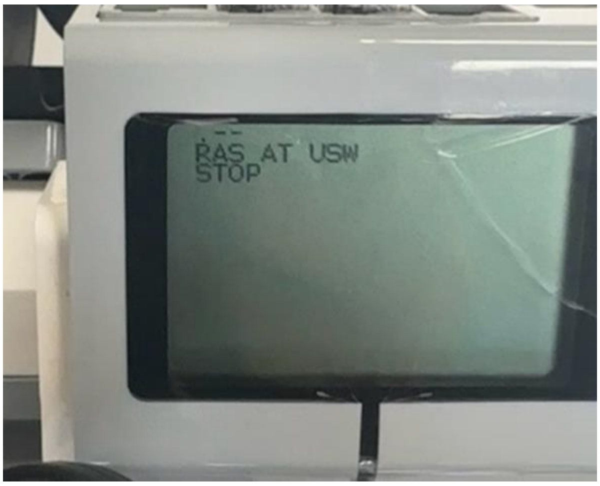

Figure 5: Example decoded output displayed on the EV3 screen

The robot detected transitions between red and white segments, measured the duration of each mark and gap, and classified them as dots, dashes, character boundaries, and word spaces. The decoded message was displayed on the EV3 screen during operation.
Colour detection remained stable under normal laboratory conditions, and integration between movement control and decoding logic operated without interruption during each test run. Small variations in sensor readings were observed across repeated runs but did not significantly affect decoding accuracy.
The ultrasonic sensor also functioned as intended by stopping the robot when an obstacle was detected within the defined distance. Overall, the system achieved the main objective of reading and displaying a Morse encoded message in real time.
A demonstration video showing the robot decoding the Morse code track during operation is provided as part of the submission.

Conclusion

This project demonstrated the successful implementation of a colour based Morse code reading system using the EV3 robot platform. After activation using the centre button on the EV3 brick, the robot moved along the printed track, detected transitions between red and white segments, and converted their duration into dots and dashes representing Morse symbols. These symbols were translated into characters and displayed on the EV3 screen during operation.

The results confirmed that timing based decoding could be applied reliably while the robot moved continuously along the track, showing effective integration between sensing, movement control, and symbol translation. The ultrasonic sensor also operated correctly by stopping the robot when an obstacle was detected ahead.

Simulation supported early development by allowing decoding behaviour to be tested before deployment on physical hardware, while verification on the real robot confirmed correct operation under laboratory conditions.

Overall, the system achieved the main objective of reading and displaying a Morse encoded message in real time using colour detection and timing measurements on the EV3 platform.

References

Afzal, A., Katz, D.S., Goues, C.L. and Timperley, C.S. (2020) ‘A Study on the Challenges of Using Robotics Simulators for Testing’. *arXiv:2004.07368 [cs]* . Available at: https://arxiv.org/abs/2004.07368.

Chukwurah, N., Adebayo, A.S. and Ajayi, O.O., 2024. Sim-to-real transfer in robotics: Addressing the gap between simulation and real-world performance. International Journal of Robotics and Simulation, 6(1), pp.89-102.

Smart, P.R. (2020) ‘Planet Braitenberg: Experiments in virtual psychology’. *Cognitive Systems Research* 64, pp. 73–95. doi: https://doi.org/10.1016/j.cogsys.2020.06.001.

Appendixes

Full program

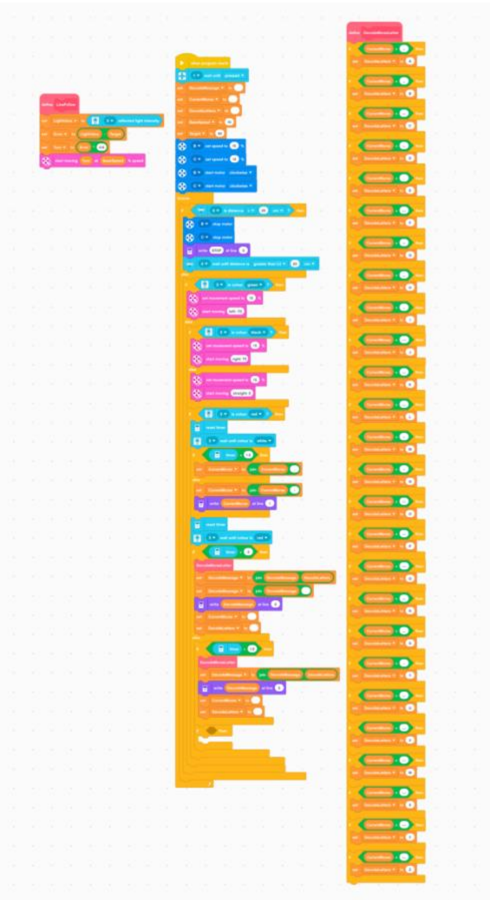

Figure 6: full program structure